Engineering Mathematics - I · Unit_02 · Unit 2

DRAFT — UE26MA141A · Unit_02 · Unit 2 · Particular integral of standard functions - Type 2

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
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M1

Particular integral of standard functions - Type 2

1 Particular integral of standard functions - Type 2

Study focus: Particular integral of standard functions - Type 2. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000078 · EV0000079

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In today's sessions, we are learning a method to solve a non-homogeneous linear differential equation of the type f of df is equal to capital X when capital X is trigonometric function of the type sine a x plus b or cos a x plus b.

Key details

- If the resultant function still contains any d term, then convert those d terms into d square terms and then replace d square by minus into a square.
- And d write as it is.
- D square replace it by minus into a square.
- And d cube write it as d square into d.
- Replace only d square by minus into a square.
- For cos a x plus b is cos e x plus b by f of d square is cos e x plus b by f of minus into a square provided the denominator f of minus a square not equal to 0.
- Now the next question is what if the denominator f of minus a into a square equal to 0.
- When b square plus a square is a factor of f of b.
- Then f of minus a square will be equal to 0.
- Sine a x plus b by f of d square is x into sine a x plus b by f dash of minus a square provided the denominator f dash of minus a square not equal to 0.
- C x has to be written outside the operator.
- x into sine a x plus b by f dash of minus a square.
- And in f dash d, you replace only d square terms by minus into a square.
- If that function still contains any d term, then convert those d terms into d square terms and then replace d square by minus a square, provided the demo denominator not equal to 0.

- It is cos a x plus b by f of d square is x into cos a x plus b by f of minus into a square where the denominator not equal to 0.
- Now the next question is what if f of minus f dash of minus a square equal to 0?
- The rule continues if f dash of minus a square equal to zero, then sin a x plus b by f of d square is x square.
- into sine a x plus b by f double dash of minus a square where in f double dash of minus a square not equal to 0.

Example or application from the lecture

- I'll just give you an example.
- Suppose say f of d, the given function of d is say d plus d square plus dq.
- In the given f of d, differentiate it with respect to d to get f dash d.

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EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 05:00–09:59

rule continues if f double if double dash of minus a square is 0 then it is x cube into sin a x plus b by f triple dash of minus a square and so on.

Key details

- But we can use the same rule for the following functions also.
- Here you have product of two trigonometric functions.
- Convert the product of trigonometric functions into sum or difference by using transformation formulae.
- See, we know that sine A into cos B.
- is 1 by 2 sin a plus b plus sin a minus b.
- Using the transformation formulae, convert the product of two functions into summary of the trigonometry functions and use the root.
- Using the trigonometric identity.
- What is sin square ax?
- It is 1 minus cos 2x by 2.
- Cos square ax 1 plus cos 2x by 2.
- Using the trigonometric identity, you just write sin square or cos square in terms of cos e.
- Ax or sine Ax and then use the same rule.
- This so same rule applicable for sine cube Ax or cos cube Ax using the same using the trigonometric identity.
- It is a non-homogeneous linear differential equation.
- Because of cos 2x plus 3.
- Non-zero function on the right hand side of the differential equation.
- It is a linear differential equation.
- What is the order of the differential equation?
- We know that.

- The general solution of this non-homogeneous differential equation has two components, the complementary function and the particular integral.
- What is the first step?
- Step one is to find the complementary function.
- So the auxiliary equation is mq plus 1 is equal to the equal to 0 because f of d here is dq plus 1 replace d by m and equate it to 0. mq plus 1 is equal to 0.
- M plus 1 into M square minus M plus 1.
- What is the formula used here?
- A Q plus B cube.
- What is AQ plus B cube?
- A plus B into A square minus A B plus B square.
- To factorize this to find the roots of this, use the quadratic formula minus b plus or minus square root of b square minus four ac by two a and find the roots.
- The roots are m is equal to minus one and half plus or minus root three by two into i.
- The first root is a real root.
- And the other two roots are complex in nature.
- For the real root.
- Alpha here is half and beta is root 3 by 2.
- It is e to the power alpha x into.
- A C2 cos beta x plus c3 psi beta x.
- 1 by 2 into x into c2 cos root 3 by 2 into x.

Example or application from the lecture

- Now let us understand this method in a simple way by taking an example.
- Consider the differential equation dq plus 1 of y is equal to cos 2x plus 3.
- Let us first analyze this differential equation.

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EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 10:00–14:59

plus C3 sin root 3 by 2 into x.

Key details

- It is 1 over f of d of cos 2x plus 3.
- by f of minus into a square provided f of minus into a square not equal to 0.
- Because a here is 2.
- Cause 2x plus 3 by C D I have written as it is, and D square is replaced by minus into 2 square.
- See please note here it is not minus 2 the whole square.
- It is minus into 2 square, which is minus 4.
- See, I told you if there is any D term remaining, D term has to be converted into D square term.
- How do we convert this 1 minus 4D into D square term?

- Multiply the numerator and denominator by its conjugate.
- What is a conjugate of 1 minus 4d?
- It is 1 plus 4d.
- Multiply the numerator and denominator with 1 plus 4d.
- See you please look here.
- See this 1 plus 4d or 1 over, 1 plus 4d are 1 over, 1 minus 4 d are all operators.
- I'll write first I'll simplify the denominator.
- What is that rule?
- We place d square by minus into 2 square which is minus 4.
- That is minus into minus become plus 64 plus 1 65.
- Cas 2x plus 3.
- What do you mean by d?
- It is the first order differential operator d by dx.
- It is minus 2 into sin 2x plus 3.
- Just look at this problem.
- A non-homogeneous linear differential equation with constant coefficients and the order of the differential equation is 4 because the equation has d square into d square, d to the power 4.
- What is d to the power 4?
- It is fourth order differential operator d4.
- 4 by dx to the power 4.
- Now, because it is a non-homogeneous linear differential equation, we need to find the complementary function and the particular integral.
- The first step is to find the complementary function.

Example or application from the lecture

- The particular integral is given by cos of 2x plus 3 divided by dq plus 1 because f of d is dq plus 1.

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EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

Core explanation · 15:00–19:59

M minus 1 the whole square into m square plus 1 equal to 0.

Key details

- How did I get this?
- You just have to replace d by m.
- And equate the expression to 0.
- That is the auxiliary equation.
- There are 4 roots and the roots are.
- 1 comma 1 comma plus r minus i.
- Why these two 1 and 1 roots?
- Because m minus 1 the whole square is 0.

- M minus 1 into m minus 1 0.
- 1 comma 1 are the two roots here.
- M square is equal to minus 1.
- M is equal to square root of minus 1, which is plus r minus i.
- The first two roots are real and equal, and the next two roots are complex in nature.
- The complementary function is c1 plus c2x into e to the power 1x.
- This is for the first two equal roots.
- Next for this complex roots.
- The second step is to find the particular integral.
- Now I have written particular integral as e power x by f of d plus car square x by 2 by f of d because it is e power x plus car square x by 2 by f of d.
- I call this as particular integral 1 and particular integral 2 and simplify them separately.
- Pi 1 plus PI2 where PI1 is a first term and P i2 is a second term cos square x by 2 by d minus 1 the whole square into d square plus 1.
- To find particular integral and what is it?
- E power x by d minus 1 the whole square into d square plus 1.
- How do I simplify this?
- I'm using rule 1.
- What is that rule 1?
- E power e x by f of d is e power a x by f of a.
- Provided f of a not equal to 0, but it's very clear from this d minus 1 the whole square 1 comma 1 are the two roots.
- If I directly substitute d is equal to 1, I get 0 in the denominator.
- So what I do is first I apply the operator 1 over d square plus 1 on e par x.
- And then I'll apply the operator 1 over d minus 1 the whole square on the resultant function of x.
- Replace b by.
- What is a here?
- That is 1 by 2.
- How do I find this now if I replace d by 1 I get f of a 0.
- If I so if I differentiate it again, again I am getting 0.
- So this is equal to first let me differentiate it one time.
- It is x into e power x by.
- 2 into 2 into d minus 1, which is 4 into d minus 1.
- Why did I write x?
- The rule says if f of e is 0, e po ax by f of d is x into e par ax by f dash d and in f dash d replace d by 1.
- X square into E power x by 4 into D minus 1.
- Differentiation is 1.

Example or application from the lecture

- See in the given differential equation f of d is d minus 1 the whole square into d square plus 1.
- Let us solve this algebraic equation because the degree of this algebraic equation is 4.

- Add the two to get the particular integral of the given differential equation.

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EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 20:00–24:59

It is ca square x by 2 by d minus 1, the whole square by d square plus 1.

Key details

- I don't have any formula for ca square x by 2 by f of d.
- That is I cannot directly find particular integral of ka square x by 2.
- I have formula for cos a x or sine x, but not for cos square x by two.
- I use trignometric identity and write cos square in terms of cos ax or cos bx.
- Cos x by 2.
- What I do now it here in the numerator it is the sum of two terms.
- I split it into two terms.
- It is 1 by 2 into d minus.
- 1 the whole square into d square plus 1 plus half into cos x by d minus 1 the whole square into d square plus 1.
- Which rule I am using?
- I'm using rule 1.
- Because 1 can be written as 1 into e power 0x.
- Replace d by 0.
- What is that rule?
- Cos a x by f of d replace d square by minus into a square.
- I apply the operator 1 over d minus 1 the whole square on cos x first.
- And I'll apply the operator d square plus 1 on the resultant function of x.
- So it is first term is 1 by 4 plus 1 by 2 into d square plus 1 into cos x by d minus 1 the whole square.
- What is it? d square minus 2d plus 1.
- How do I simplify this?
- replace d square by minus into a square.
- D term right as it is.
- So what I do is first I simplify the term inside the bracket and then apply the operator 1 over d square plus 1.
- It is integration of cos x with respect to x.
- without writing the constant of integration.
- It is sine x.
- How do I calculate this?
- I use rule sin a x plus b by f of d replace d square by minus into a square I get minus 1 square plus 1 which is 0.
- When you are writing x here you have to differentiate the denominator.
- Differentiation of d square is 2d.

- So for this I need not need not have to use any rule because 1 over d of sin x, that is integration of sine x with respect to x, which is minus cos x, so it is 1 by 4 plus x into cos x by 8.
- It is particular integral 1 plus particular integral 2 and.
- The general solution y is equal to the complementary function plus particular integral.
- See if they ask you to find only the particular integral, you can write up till here.
- If they ask you to solve the differential equation, you should mention this.
- Y is equal to complementary.
- function plus particular integral.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Particular integral of standard functions - Type 2 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_of_standard_functions_-_Type_2__d25d020162de0016.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 2 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_of_standard_functions_-_Type_2__d25d020162de0016.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral of standard functions - Type 2 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_of_standard_functions_-_Type_2__d25d020162de0016.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 2 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_of_standard_functions_-_Type_2__d25d020162de0016.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00018	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00019	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00020	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00021	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

C Revision and retrieval practice